

Intelligent Complaint Classification Using Localized Transformer Architectures (E-Commerce): Impact statement

This project has the potential to improve customer service in the e-commerce sector by making complaint classification faster, more consistent, and more responsive to the actual language used by customers. The proposed transformer-based system will automatically classify customer complaints into relevant categories, enabling them to be routed to the appropriate support function with less reliance on manual triage. This can reduce repetitive workload for customer-support personnel, improve operational efficiency, and help customers receive attention from the teams best equipped to address their concerns.

The project's potential social impact extends beyond efficiency. By allowing customers to communicate their problems naturally through text rather than relying solely on predefined options or keywords, the system can make complaint submission and routing more accessible. More accurate classification can also promote consistency in how complaints are handled, supporting fairness and improving the overall customer experience. These benefits are particularly relevant as e-commerce businesses manage increasing volumes of complaints across websites, email, live chat, and mobile applications.

Fairness and responsible AI are central to the project's design. The dataset brings together publicly available consumer complaint and customer-support data to provide broader coverage of complaint types and language patterns. The model will be evaluated not only on overall performance but also across individual complaint categories, including less-represented categories. Stratified data splitting, class weighting, and metrics such as precision, recall, macro F1, and per-class F1 will help identify and address differences in performance. This approach supports the project's commitment to fairness and transparency.

The system is designed as a decision-support tool rather than a replacement for customer-support personnel. Human oversight will remain important for uncertain or sensitive complaints, allowing organizations to combine the efficiency of automated classification with human judgement. This supports accountability while preserving customer dignity and ensuring that important decisions are not made solely by an automated system.

Over time, successful implementation could provide e-commerce organizations with a scalable approach to managing large volumes of customer complaints while generating more consistent and efficient support workflows. The project's use of publicly available data also provides a practical foundation for research without requiring access to private company records.

While performance may vary across businesses, complaint types, and customer populations, documenting these limitations and continuously evaluating the model will support responsible improvement. Overall, the project aims to demonstrate how localized transformer architectures can be applied to a practical customer-service challenge while advancing efficiency, accessibility, fairness, transparency, accountability, and customer satisfaction.